

Python Basics Quiz

Topics: Numbers, Strings, Lists, Tuples, Input/Output

Name: _____ Date: _____

Section A: Multiple Choice Questions

1. What is the result of `2 ** 3`?
 - a) 6
 - b) 8
 - c) 9
 - d) 5
2. Which of these is NOT a valid variable name in Python?
 - a) `_value`
 - b) `value1`
 - c) `1value`
 - d) `Value_1`
3. What is the output of `print("5" + "5")`?
 - a) 10
 - b) 55
 - c) Error
 - d) 5 5
4. What does `"Hello".upper()` return?
 - a) hello
 - b) HELLO
 - c) Hello
 - d) Error
5. Which of the following creates a list?
 - a) `(1, 2, 3)`
 - b) `{1, 2, 3}`
 - c) `[1, 2, 3]`
 - d) `<1, 2, 3>`
6. What is the output of `type((1, 2, 3))`?
 - a)
 - b)
 - c)
 - d)
7. What does `int("25")` return?
 - a) "25"

- b) 25
- c) 25.0
- d) Error

8. What is the output of `print(3 == 3.0)`?

- a) False
- b) True
- c) Error
- d) None

9. Which method converts all characters of a string to lowercase?

- a) `.lower()`
- b) `.small()`
- c) `.down()`
- d) `.minimize()`

10. What will `[10, 20, 30].index(20)` return?

- a) 0
- b) 1
- c) 2
- d) Error

11. What is the data type of the result of `input()`?

- a) int
- b) float
- c) str
- d) Depends on what user enters

12. Which of the following is used to join two tuples?

- a) +
- b) -
- c) *
- d) /

13. What is the output of `len("Hello World")`?

- a) 10
- b) 11
- c) 12
- d) 13

14. What is the output of `print(10 % 3)`?

- a) 3
- b) 1
- c) 0
- d) 3.33

15. Which of these correctly formats output using an f-string?

- a) `print(f"My name is {name}")`

- b) `print("My name is" + {name})`
- c) `print(fstring(name))`
- d) `print(%name)`

Section B: Fill in the Blanks

1. ``print(type(3.14))`` will output _____.
2. The function used to convert a value to a string is _____.
3. In Python, indexing starts from _____.
4. ``"Hello"[1]`` will return _____.
5. To find the length of a list, we use the _____ function.
6. Tuples are defined using _____ brackets.
7. The ``+`` operator when used with two strings performs _____.
8. ``list.sort()`` arranges elements in _____ order by default.
9. The ``input()`` function displays a _____ message to the user if provided as an argument.
10. ``5 / 2`` gives a result of type _____, while ``5 // 2`` gives a result of type _____.
11. To convert user input into an integer, we write ``int(_____)``.
12. A tuple with only one element must include a _____ after the value, e.g., ``(5,)``.
13. The negative index ``-1`` refers to the _____ element of a list or string.
14. ``"abc" * 3`` will output _____.
15. To add multiple items to a list at once, we use the _____ method.

Section C: True or False

1. ``print()`` automatically adds a new line after printing.
2. Lists can contain elements of different data types.
3. ``int("abc")`` will run successfully and return ``0``.
4. Tuples are faster than lists for fixed data.
5. String concatenation can be done using the ``+`` operator.
6. ``5 != 5`` evaluates to ``True``.
7. ``input()`` can accept a prompt message as an argument.
8. In Python, ``//`` always rounds the result up.
9. Lists use parentheses ``()`` for declaration.
10. A tuple can be converted into a list using ``list()``.

Section D: Predict the Output

O1

```
a = 7
b = 2
print(a ** b)
print(a % b)
print(a / b)
```

Output: _____

O2

```
s = "Programming"
print(s[3:7])
print(s[:4])
print(s[5:])
```

Output: _____

O3

```
nums = [5, 10, 15, 20]
print(nums[-1])
print(nums[-2])
nums[0] = 100
print(nums)
```

Output: _____

O4

```
t1 = (1, 2)
t2 = (3, 4)
print(t1 + t2)
print(t1 * 2)
```

Output: _____

O5

```
x = "10"
y = 20
print(int(x) + y)
```

Output: _____

O6

```
fruits = ["apple", "banana", "cherry"]
print(len(fruits))
fruits.append("mango")
print(fruits)
```

Output: _____

O7

```
name = "Alice"  
age = 25  
print(f"{name} is {age} years old")
```

Output: _____

O8

```
a, b, c = 1, 2, 3  
print(a + b + c)
```

Output: _____

O9

```
t = (10, 20, 30, 40)  
print(t[1:3])  
print(t[::-1])
```

Output: _____

O10

```
word = "python"  
print(word.upper())  
print(word.capitalize())  
print(len(word))
```

Output: _____